

gdTAI V4.2 T/NK-restricted reintegration

Generated 2026-08-17. This is a development-sidecar analysis, not a classifier result or model promotion.

Refined T/NK pool: 3,927,924 of 4,023,462 cells retained (97.63%); 95,538 cells removed. scVI used 4,000 features and GPU only.

1. What changed

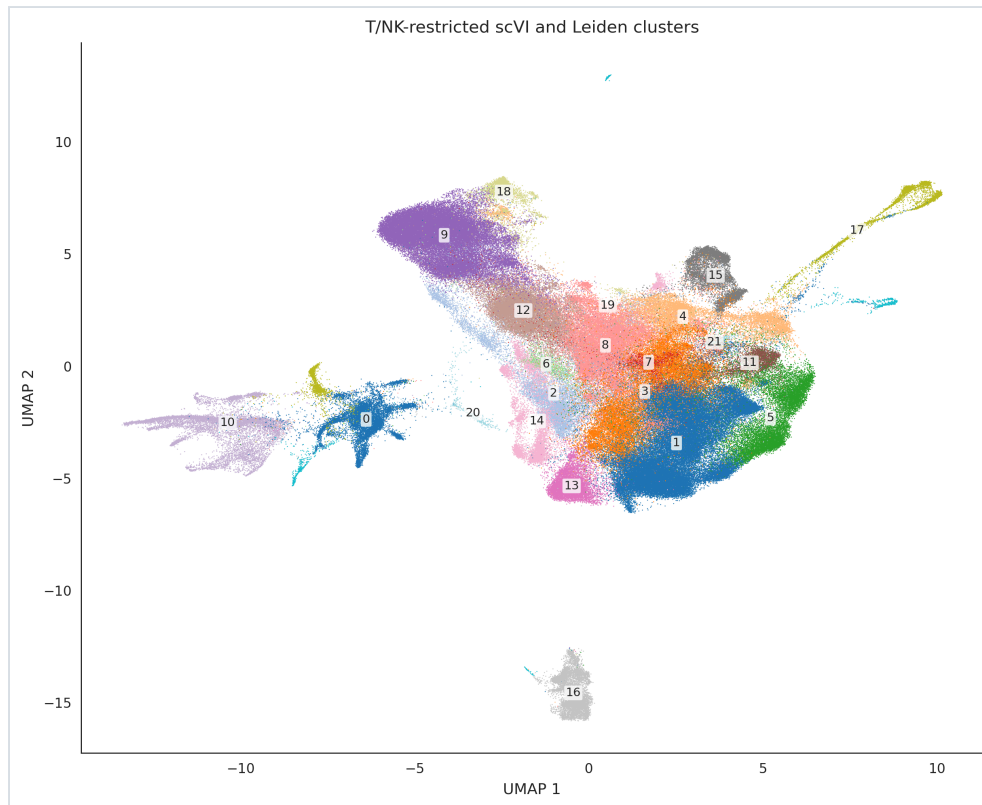
The inputs were already high-recall T/NK candidate objects, but they retained ambiguous off-target cells. This second pass preserves all non-doublet productive-TCR cells and primary NK anchors, then requires direct T-lineage, gamma-delta, NK-receptor, or multigene NK-adaptor-plus-cytotoxic evidence. Cells with broad myeloid evidence and no direct T/gdT/NK-receptor support are removed.

TRDC+ / delta-V- / weak-CD3 is not used as an NK label. It is retained as a phenotype for unsupervised review. Dropout, ambient RNA, and true NK biology remain competing explanations.

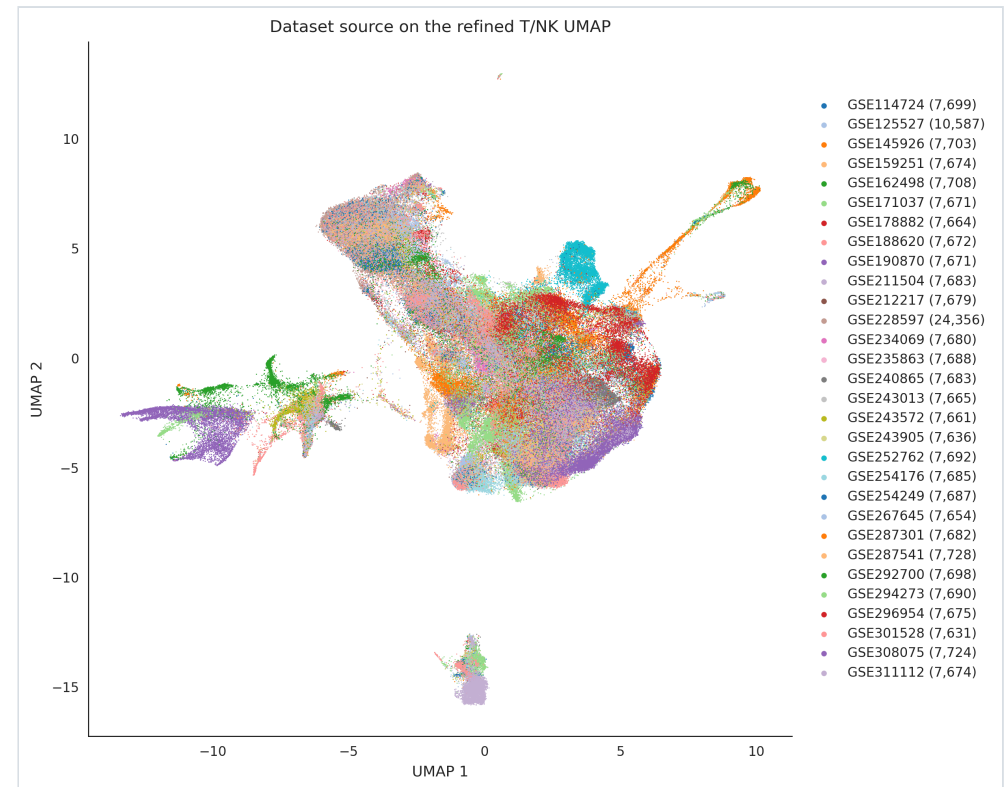
tnk_gate_keep	tnk_gate_reason	n_cells
False	removed_myeloid_conflict	28195
False	removed_no_direct_tnk_evidence	67343
True	gd_lineage	31730
True	nk_receptor	75573
True	primary_nk_protected	21054
True	productive_tcr_protected	967149
True	shared_cytotoxic_plus_nk_adaptor	9004
True	t_lineage	2823414

2. Recomputed HVGs and integration

HVGs were recomputed only after T/NK subsetting with source-balanced sampling. TCR V/J/D genes were excluded from the HVG ranking to limit clonotype-driven structure. A fixed 27-gene CD3/TCR-constant and T/NK lineage panel was forced into the 4,000-feature set. The fit completed in 25.5 minutes on NVIDIA A100 80GB PCIe.

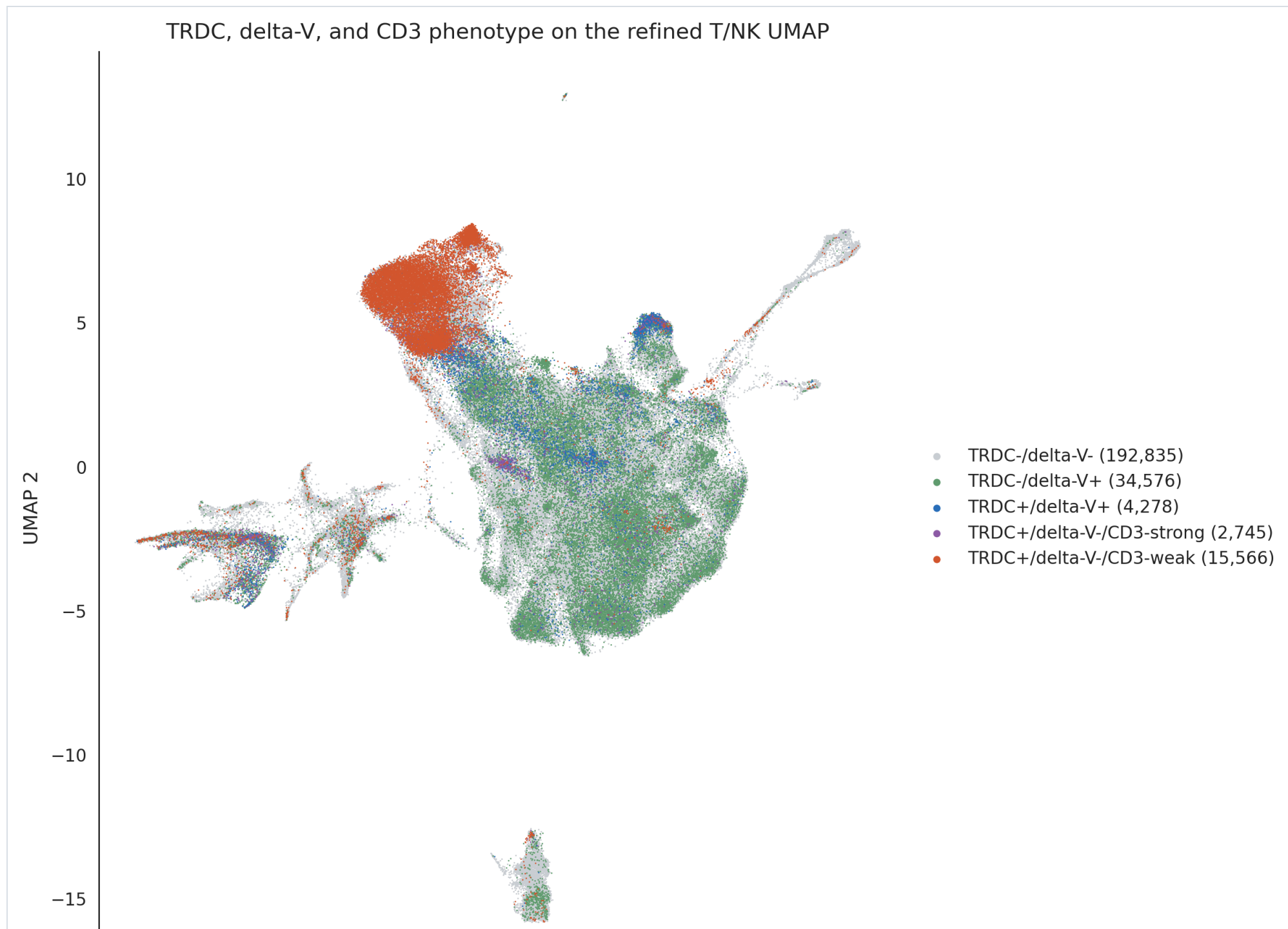


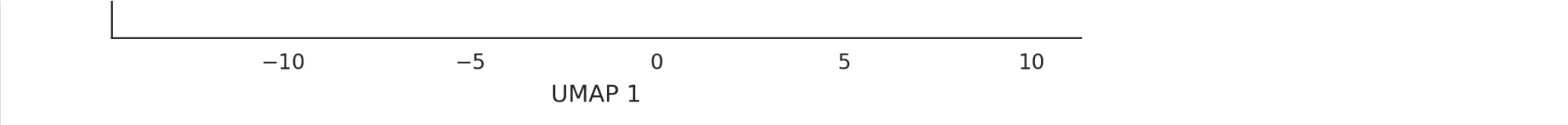
Unsupervised refined clusters.



Dataset mixing remains visible and must be considered in annotation.

3. TRDC, delta-V, and CD3 phenotype

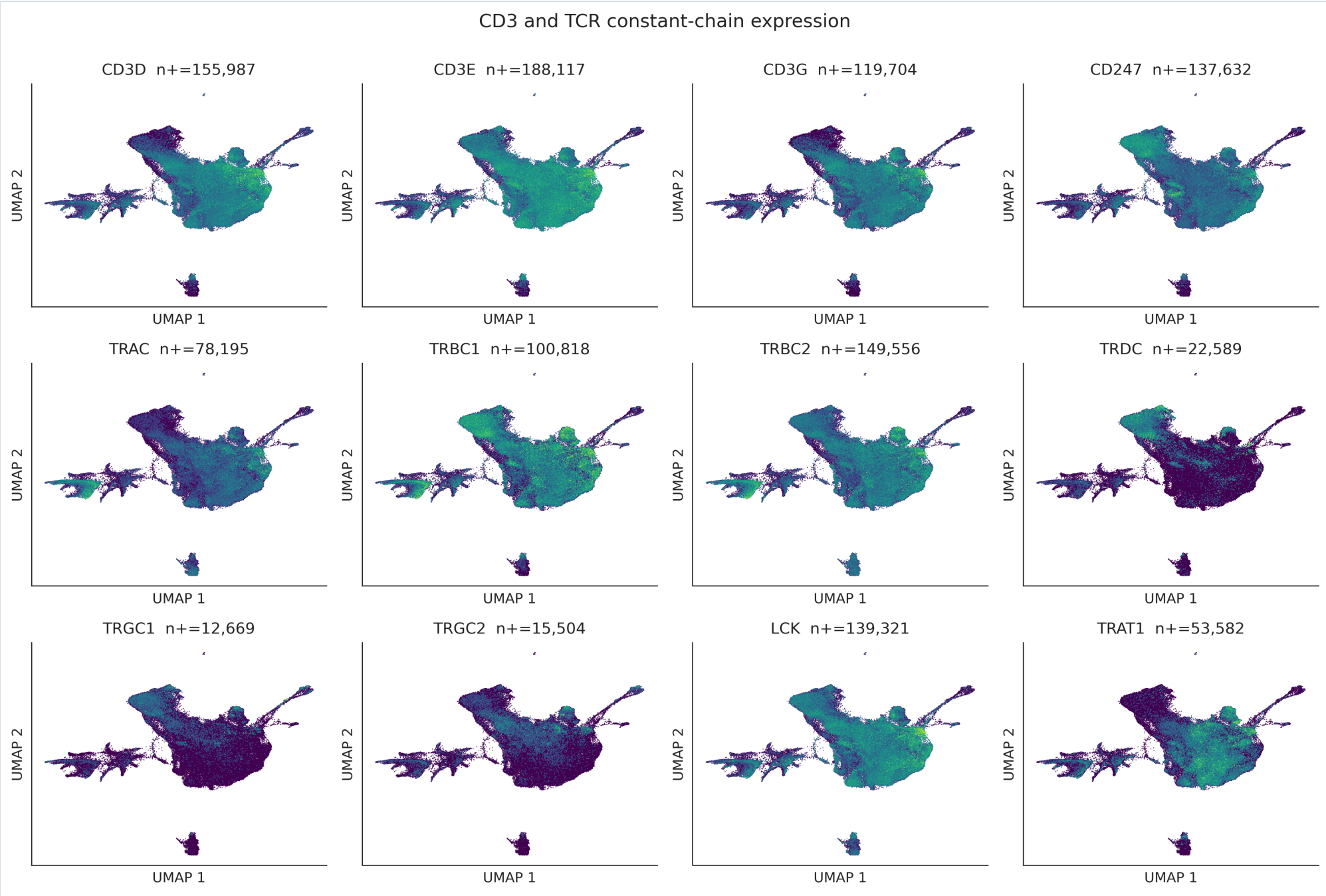


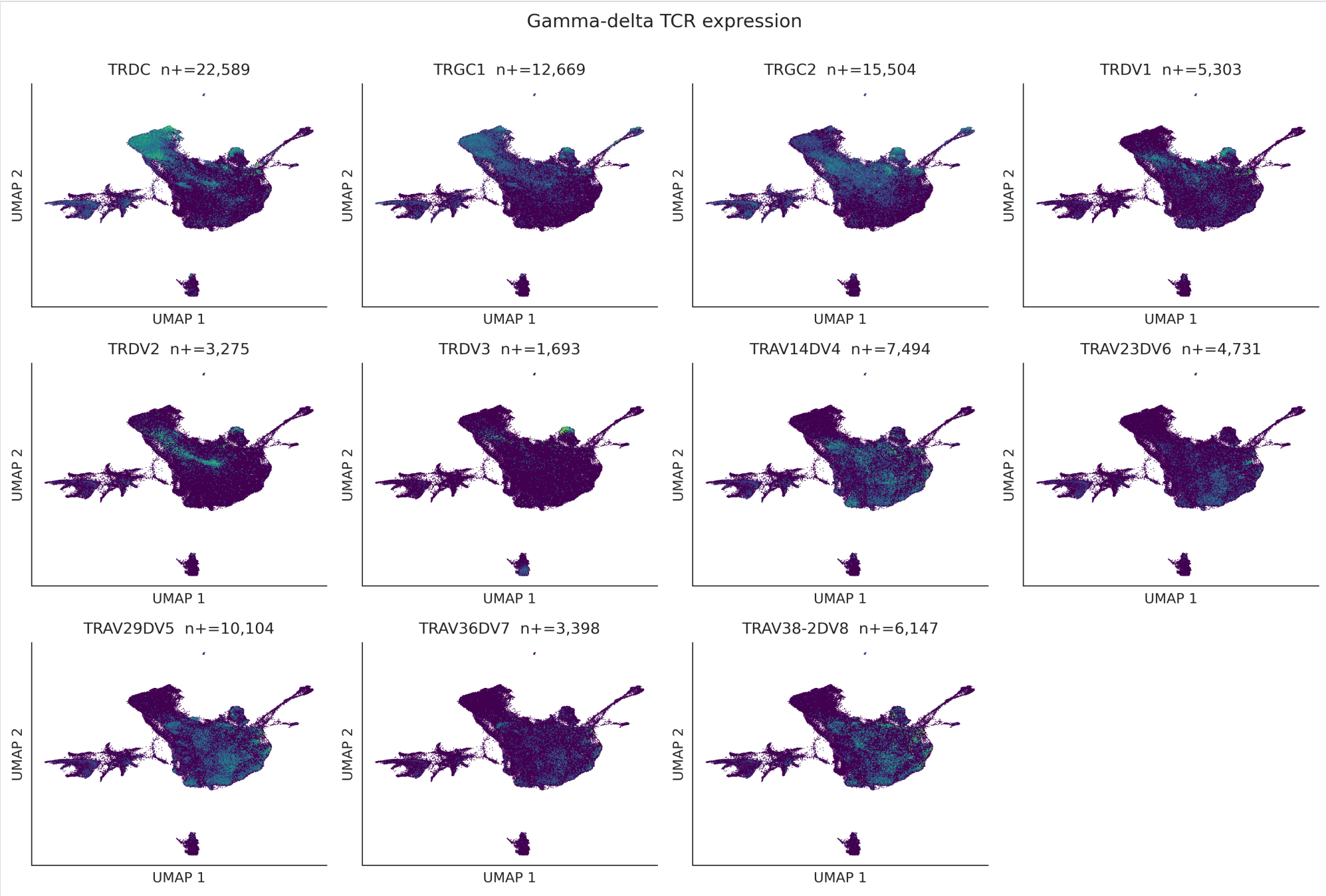


The proposed NK-like phenotype is displayed but not treated as truth.

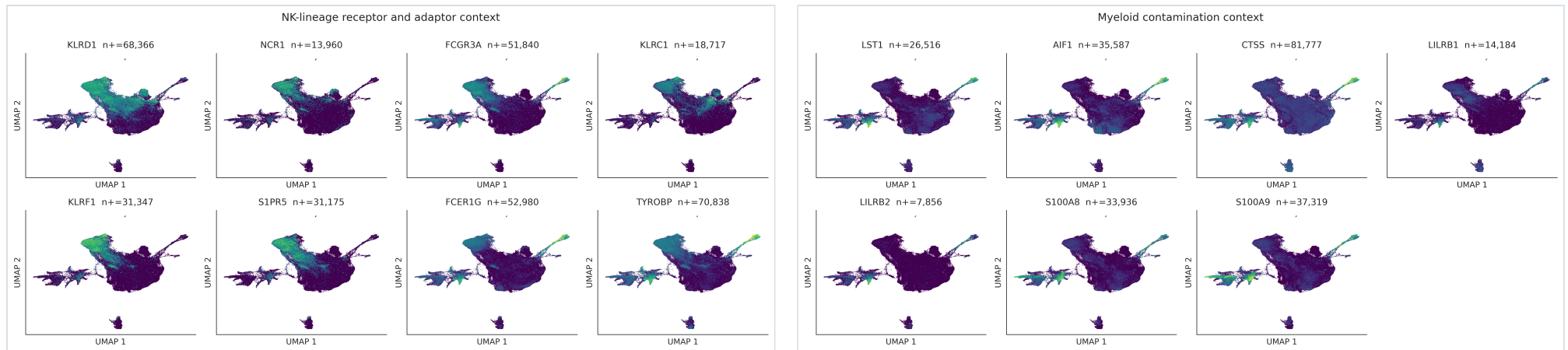
trdc_delta_v_cd3_phenotype	tnk_gate_keep	n_cells
TRDC+/delta-V+	True	67672
TRDC+/delta-V-/CD3-strong	True	43942
TRDC+/delta-V-/CD3-weak	True	178674
TRDC-/delta-V+	True	585051
TRDC-/delta-V-	False	95538
TRDC-/delta-V-	True	3052585

4. CD3 and TCR genes

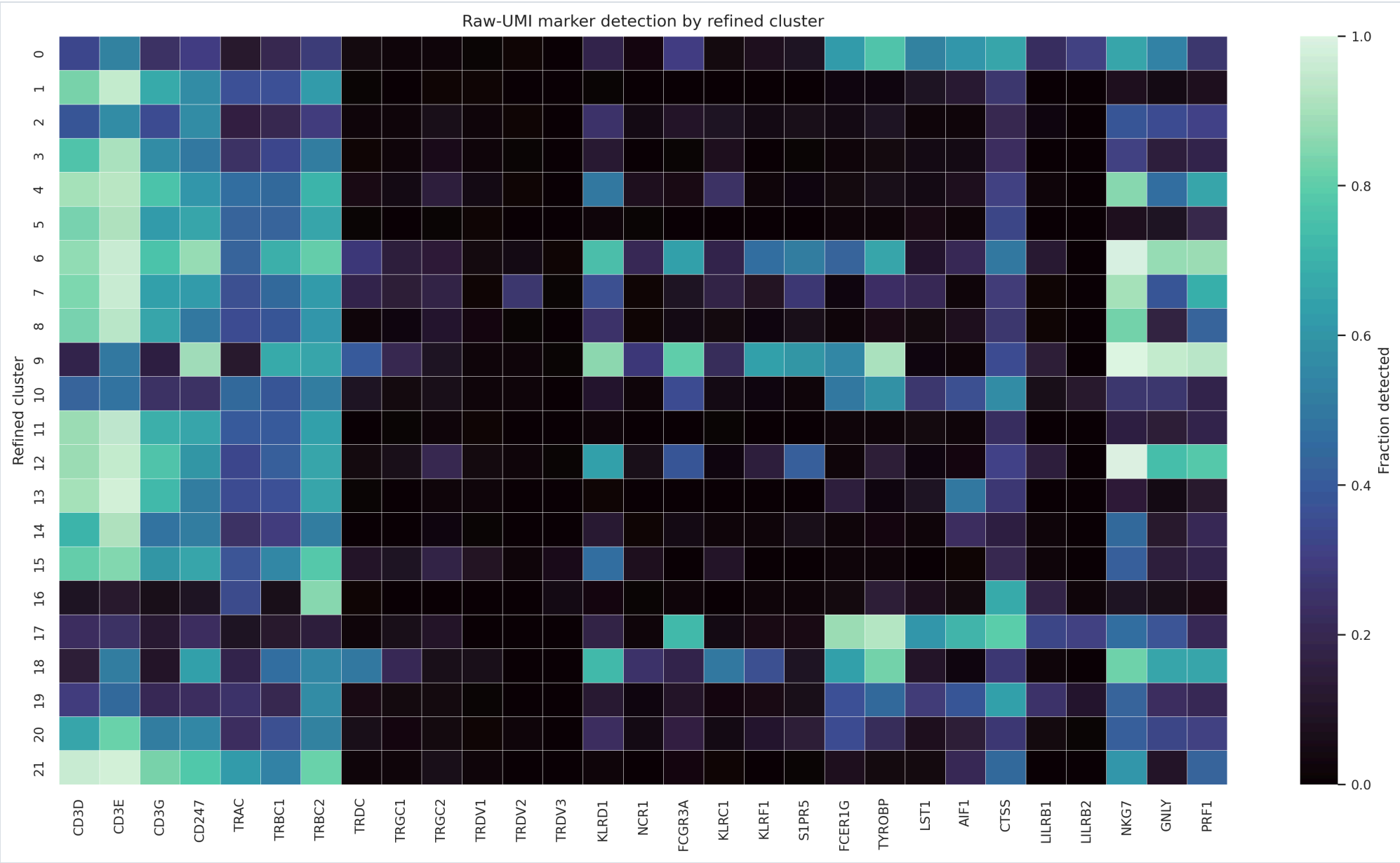




5. NK and contamination context



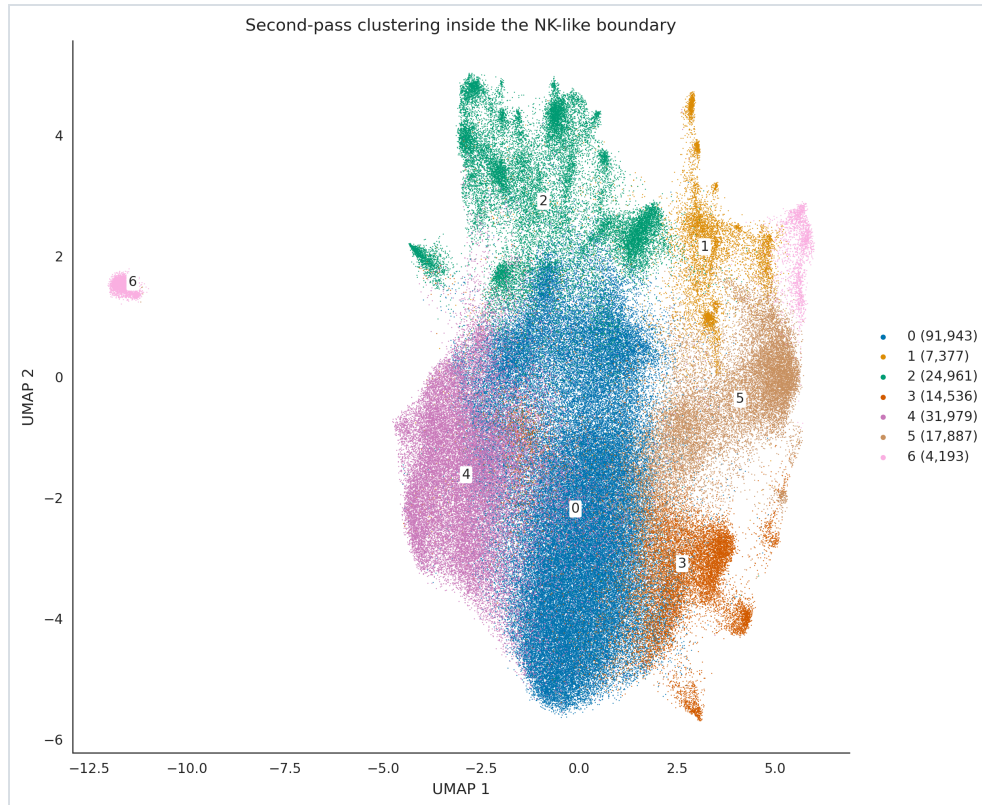
6. Cluster-level evidence



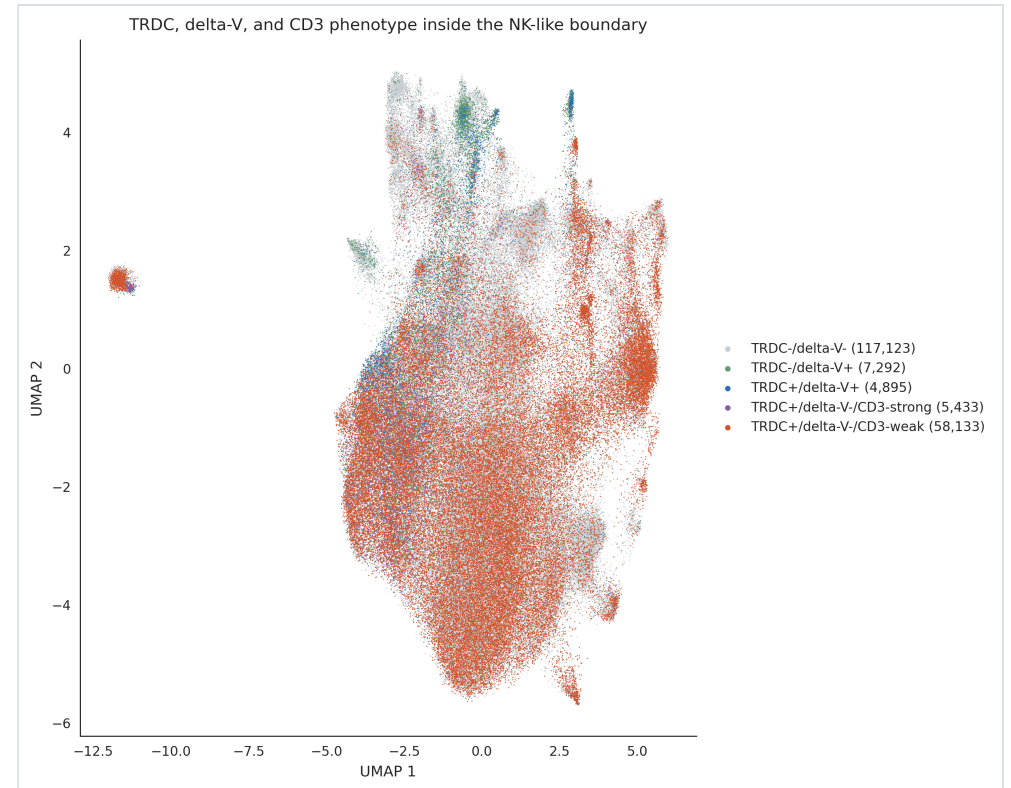
cluster	n_cells	n_primary_nk	n_productive_tcr	primary_nk_fraction	productive_tcr_fraction
9	406026	19227	7894	0.0474	0.0194
18	69927	1148	1354	0.0164	0.0194
2	223236	320	9578	0.0014	0.0429
6	33418	21	14178	0.0006	0.4243
12	295726	174	102309	0.0006	0.3460
4	224352	128	38595	0.0006	0.1720
20	9737	5	4723	0.0005	0.4851
7	69952	13	17636	0.0002	0.2521
16	108584	7	5584	0.0001	0.0514
0	169253	6	28408	0.0000	0.1678
8	419801	5	108993	0.0000	0.2596
5	284518	0	84517	0.0000	0.2971
1	751551	0	365763	0.0000	0.4867
3	432234	0	59380	0.0000	0.1374
13	124524	0	42509	0.0000	0.3414
11	105635	0	18144	0.0000	0.1718
10	37433	0	2997	0.0000	0.0801
14	66044	0	49321	0.0000	0.7468
17	33987	0	1556	0.0000	0.0458
15	45839	0	376	0.0000	0.0082
19	11753	0	1448	0.0000	0.1232
21	4394	0	1886	0.0000	0.4292

7. Second-pass NK-boundary clustering

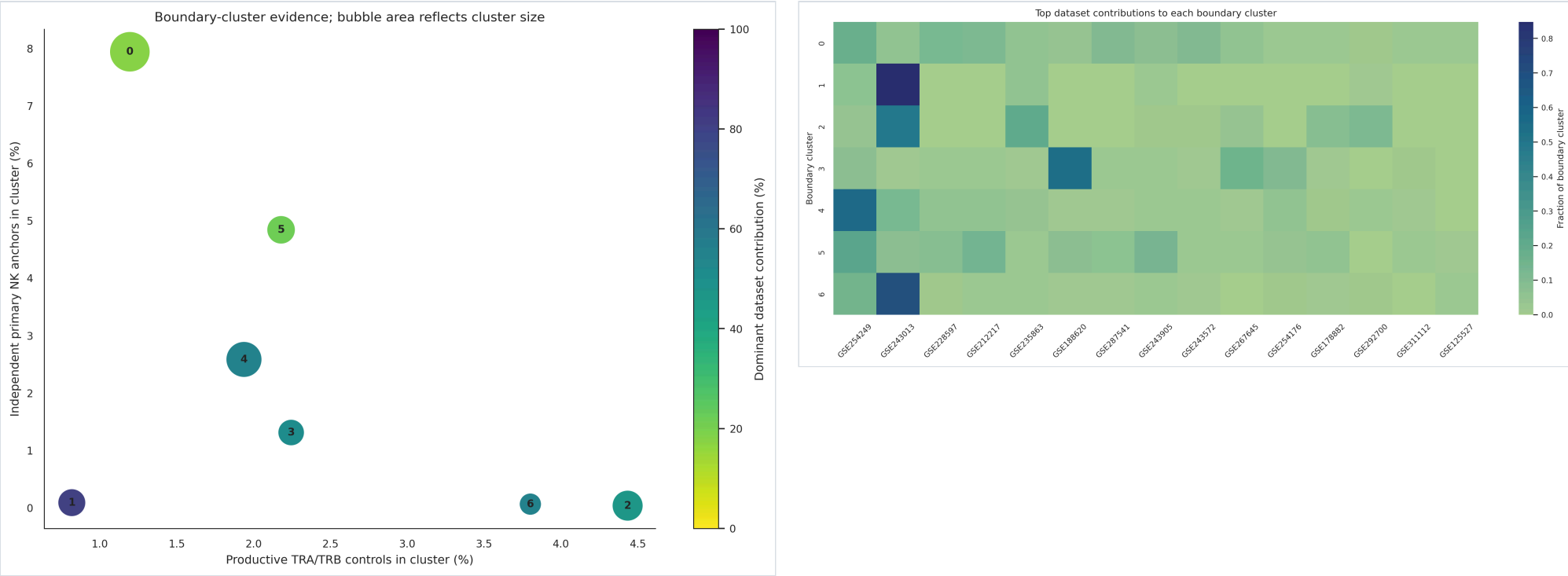
Refined clusters 9 and 18 contain 475,953 cells and recover 20,375/21,054 independent primary NK anchors (96.77%). They also contain 9,248 productive TRA/TRB controls, so neither parent cluster is labeled wholesale. Nine GPU Leiden runs were performed inside this boundary; the frozen review partition `boundary_r0.4_s29` contains 7 subclusters. At resolution 0.4, the mean adjusted Rand index across three seed pairs is 0.962 (minimum 0.952), supporting the reproducibility of this review partition.



Second-pass unsupervised subclusters.



The weak-CD3 phenotype is enriched but is not an NK label.



resolution	n_seed_pairs	mean_adjusted_rand_index	min_adjusted_rand_index	max_adjusted_rand_index
0.2000	3	0.9399	0.9115	0.9959
0.4000	3	0.9617	0.9521	0.9671
0.8000	3	0.8450	0.7852	0.9055

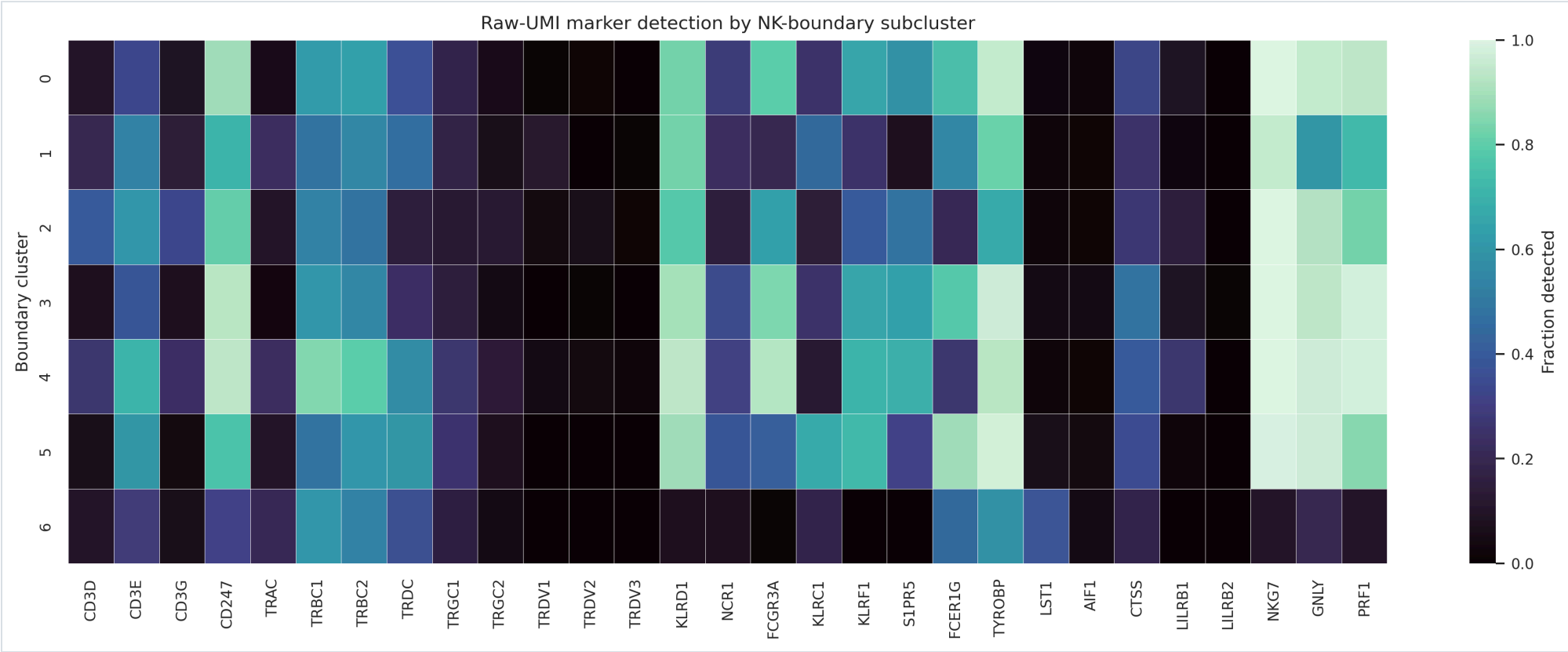
8. Review-level NK definition

Boundary clusters 0, 3, 5 form the strongest review-level NK core: weak `CD3D/CD3G` , high NK receptors, `FCER1G/TYR0BP` , and cytotoxic genes, low `LST1/AIF1` , and representation across 24-29 datasets. Together they contain 257,569 cells, 17,385 independent NK anchors (82.57% of all primary anchors), and 3,759 productive TRA/TRB controls (1.46%). These are review annotations, not classifier labels.

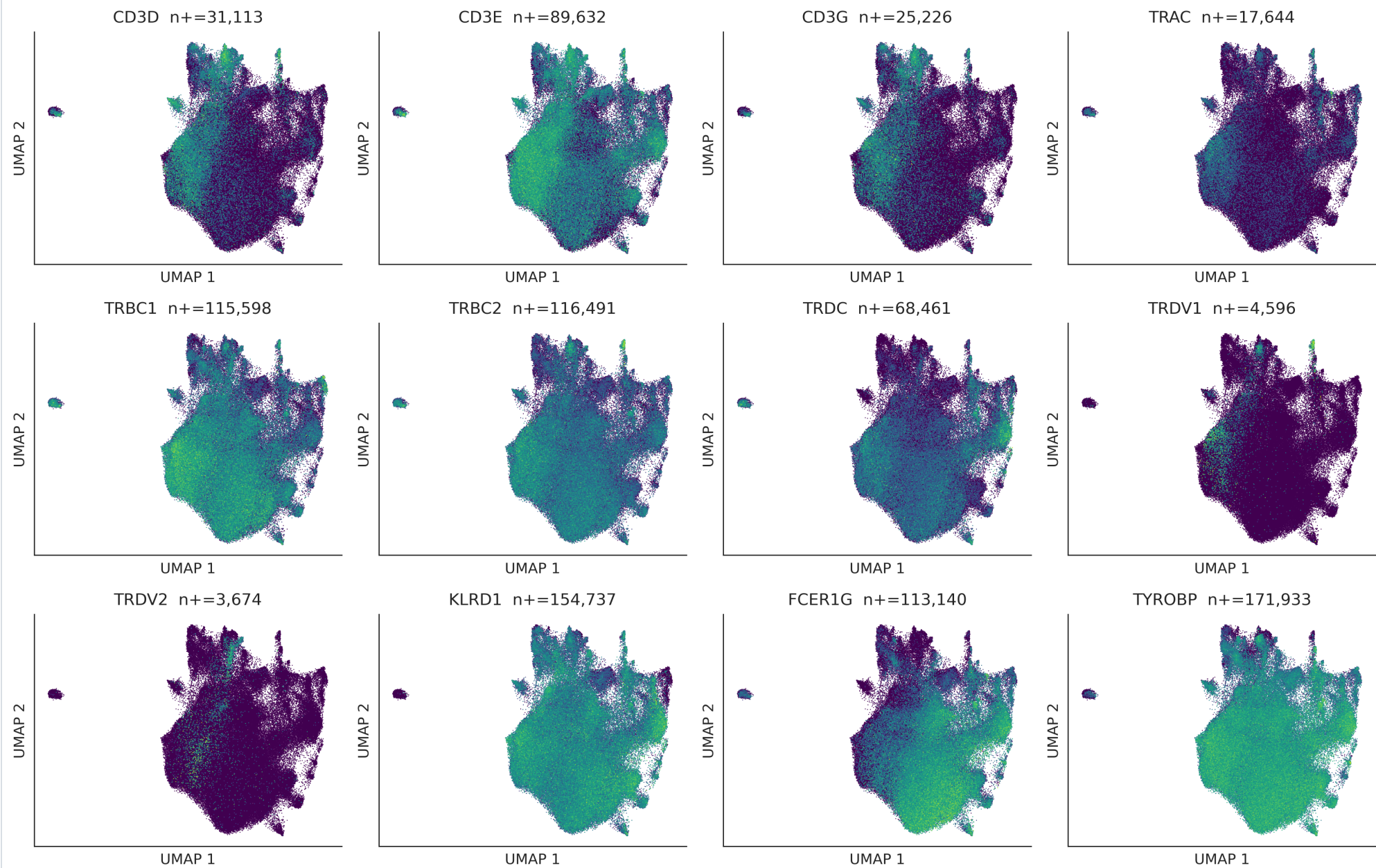
Do not use a single gene as the NK rule. `TRDC` , cytotoxic genes, `FCER1G` , and `TYR0BP` can occur in gamma-delta T cells or through ambient RNA. The core designation requires concordant cluster-level evidence and source reproducibility. Source-dominated cluster 1,

mixed T/NK cluster 4, cytotoxic-T-like cluster 2, and off-target/low-quality cluster 6 remain outside the core.

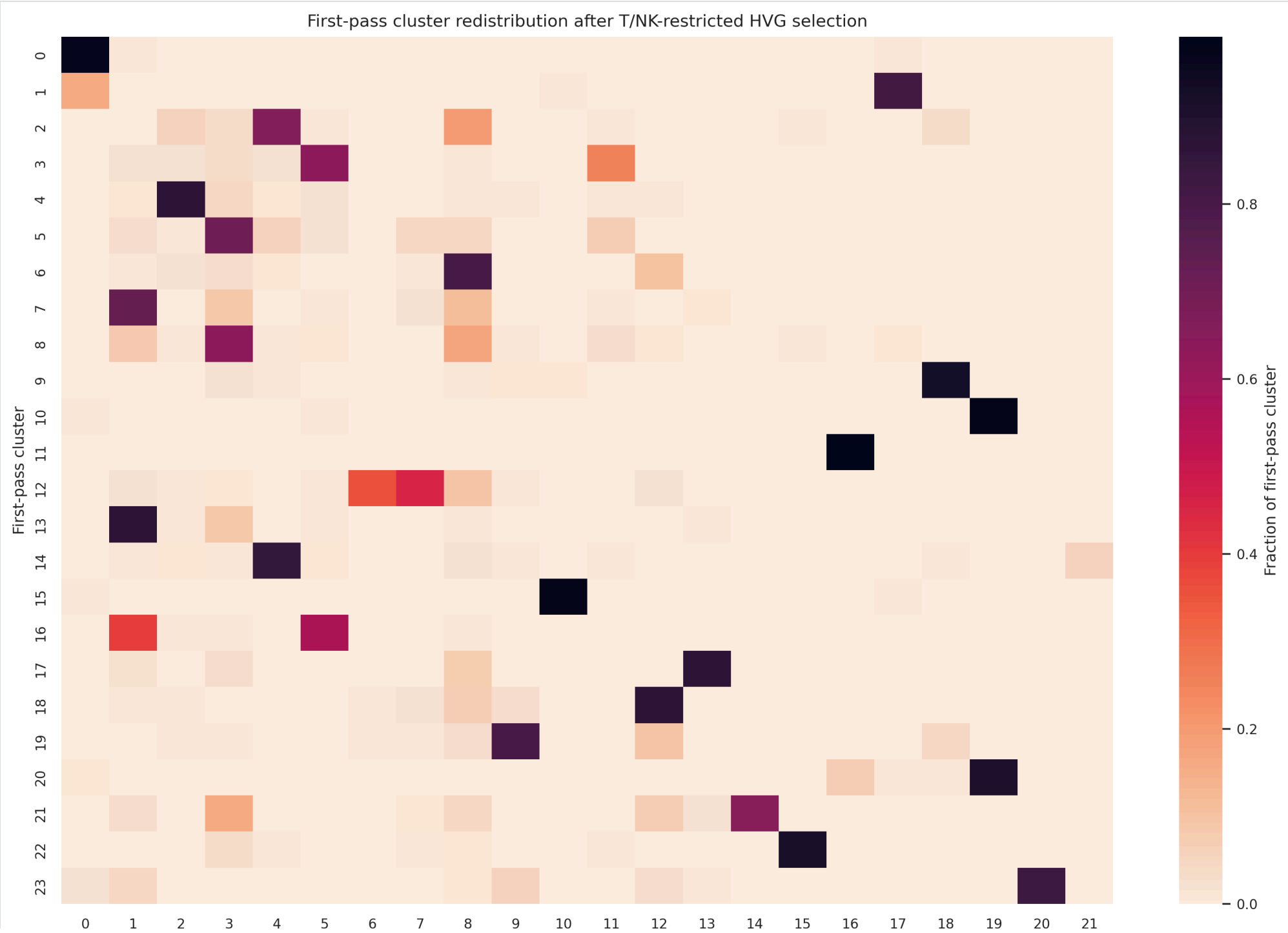
boundary_cluster	review_annotation	n_cells	n_primary_nk	n_productive_tcr	n_sources	dominant_source	primary_nk_fraction	productive_tcr_fraction	dominant_source_fraction
0	core_NK_like	190265	15112	2274	29	GSE254249	0.0794	0.0120	0.1733
1	source_dominated_NK_like_review	35596	31	291	27	GSE243013	0.0009	0.0082	0.8048
2	cytotoxic_T_like	57804	20	2563	26	GSE243013	0.0003	0.0443	0.4622
3	core_NK_like	27887	365	626	24	GSE188620	0.0131	0.0224	0.5163
4	mixed_T_NK	113540	2932	2200	29	GSE254249	0.0258	0.0194	0.5457
5	core_NK_like	39417	1908	859	29	GSE254249	0.0484	0.0218	0.2211
6	off_target_or_low_quality	11444	7	435	27	GSE243013	0.0006	0.0380	0.5557



T, gamma-delta, and NK evidence inside the NK-like boundary



9. Difference from the first pass



Refined cluster

Rows are first-pass clusters and columns are refined clusters; values are row fractions.

10. Retained cells by source

source_gse_id	tnk_gate_keep	n_cells	fraction_of_source
GSE243013	True	748188	0.9852
GSE254249	True	397778	0.9786
GSE243905	True	347177	0.9907
GSE287301	True	321965	0.9463
GSE235863	True	237023	0.9804
GSE212217	True	211187	0.9882
GSE243572	True	166787	0.8861
GSE228597	True	149198	0.9959
GSE188620	True	119093	0.9665
GSE308075	True	111333	0.9911
GSE287541	True	108777	0.9811
GSE254176	True	95796	0.9881
GSE267645	True	90323	0.9765
GSE311112	True	89992	0.9933
GSE296954	True	86360	0.9971
GSE292700	True	76208	0.9739
GSE162498	True	73904	0.9740
GSE159251	True	68792	0.9985
GSE178882	True	64856	0.9572
GSE294273	True	55403	0.9939
GSE125527	True	45529	0.9813
GSE252762	True	45003	0.9971
GSE301528	True	41996	0.9522
GSE211504	True	36684	0.9850
GSE171037	True	31721	0.9932
GSE114724	True	28651	1.0000
GSE234069	True	20639	0.9919
GSE240865	True	20563	0.9485
GSE145926	True	20292	0.9645
GSE190870	True	16706	0.9972

11. Interpretation boundary

This analysis can identify coherent NK-like and T-like transcriptomic neighborhoods, but it cannot by itself prove that every TRDC-positive, delta-V-negative cell is NK. Final NK training labels require agreement among unsupervised cluster identity, independent author/scVI NK anchors, low productive-TCR contamination, source reproducibility, and source/library ambient and doublet audits.

Reproducibility

- Config SHA-256: `dc9d68bc31ef4bb212c4bc39d7796b6cf4d07c59917f3f15d5769253ea02498e`
- Script SHA-256: `0978389725f577a2afb913c9d424b84fac38b15340060371bb4c03cbcd25ec4a`
- Staged H5AD SHA-256: `de69f8fa73e16dc88630850b73763a1e98ae3dbebcbf6a5e3884be8def740e16`
- Latent SHA-256: `2db605d8c10add04d81902e3fb586767e4485fa073d89fc565502ca75328baa0`
- Partitions SHA-256: `ae40cc6933419efd9796cc56704478a3d9563be4aa1c4849ab02824125d8e2a8`
- Boundary partitions SHA-256: `b4137a3aad812ddc2f87eed095ee4c662f2d81297e331fd93c5261c7ee930b14`
- Source H5AD mutation: **none**; classifier fitting: **none**; GitHub push: **none**.